

1.
 - (a) 9
 - (b) 4
 - (c) 9
 - (d) 0

2.
 - (a) $f : \mathbb{R} \rightarrow \mathbb{R}$
 - (b) $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ (Note the function is not defined at $x = 0$)
 - (c) $f : \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$ (Square root not defined for negative numbers)

3.
 - (a) 0
 - (b) 1
 - (c) $n\pi, \forall n \in \mathbb{Z}$